ITT06105 • DESKTOP PUBLISHING

PRINTING DESKTOP PUBLISHING DOCUMENTS

Lecture 09 | Ordinary Diploma in IT

PDF proof → printer settings → test print → quality checks

Academic year 2026/2027

Learning outcomes

After this lecture, you should be able to explain and demonstrate:

1	**Compare**	Describe basic inkjet, laser and commercial print options.
2	**Prepare**	Select suitable A4 paper, size, scaling and print settings.
3	**Explain**	Describe PDF proof, trim, safe area and bleed.
4	**Calculate**	Determine image PPI and simple printing costs.
5	**Diagnose**	Identify common defects and apply safe first checks.

COURSE CONNECTION

Why printing is part of DTP

A design is used by real people, often beyond the screen.

The final output matters

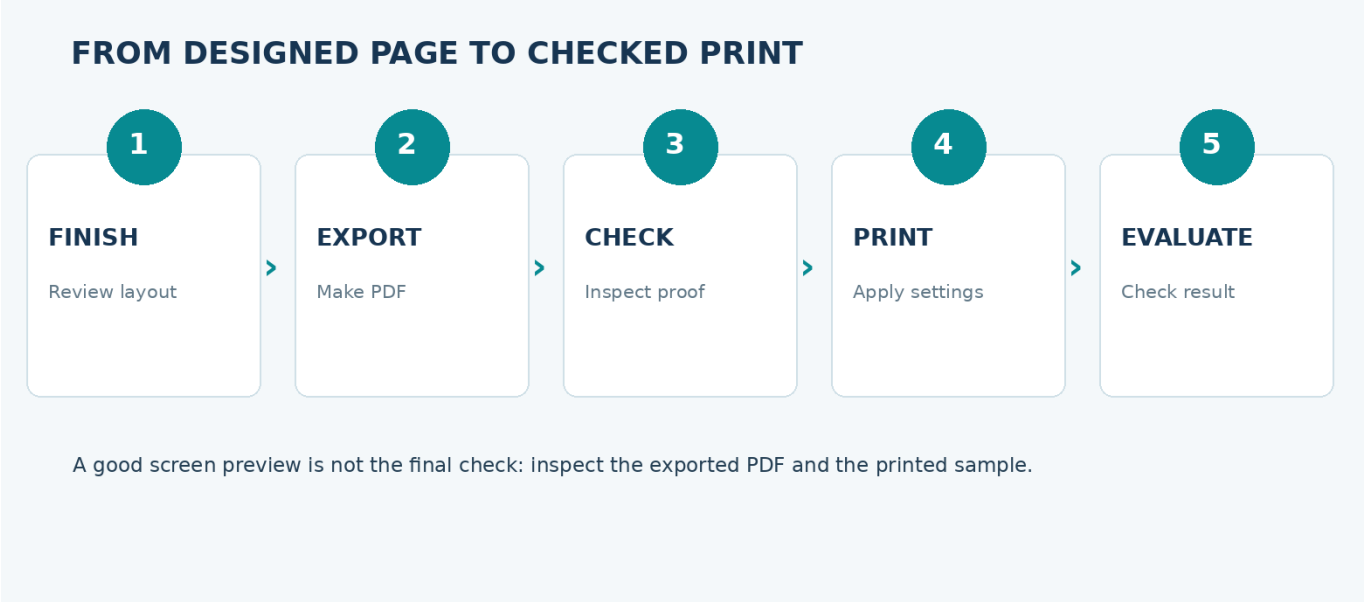
Desktop publishing does not finish when a student saves a file. For printed publications, the physical page must preserve the words, graphics, arrangement and intended quality of the design. Poor settings can make a correct layout unreadable or inaccurately sized.

Our example is the A4 “Stay Safe Online” hand-out completed in Lectures 7 and 8.

ILLUSTRATION

From designed PDF to printed hand-out

The workflow is short, but each check has a distinct purpose.



Five-stage workflow

Review the editable design; export a fresh PDF; inspect the proof; select printer settings; then examine the first printed copy. If you find an error, correct the source or settings before printing the whole batch.

EXPLAIN

Printed output and digital output

The same publication may be delivered in more than one way.

Digital publication

A PDF shared by phone, email or a learning portal is viewed on a screen. File size, screen legibility and device compatibility matter. Digital viewing does not confirm that colours or margins will print identically.

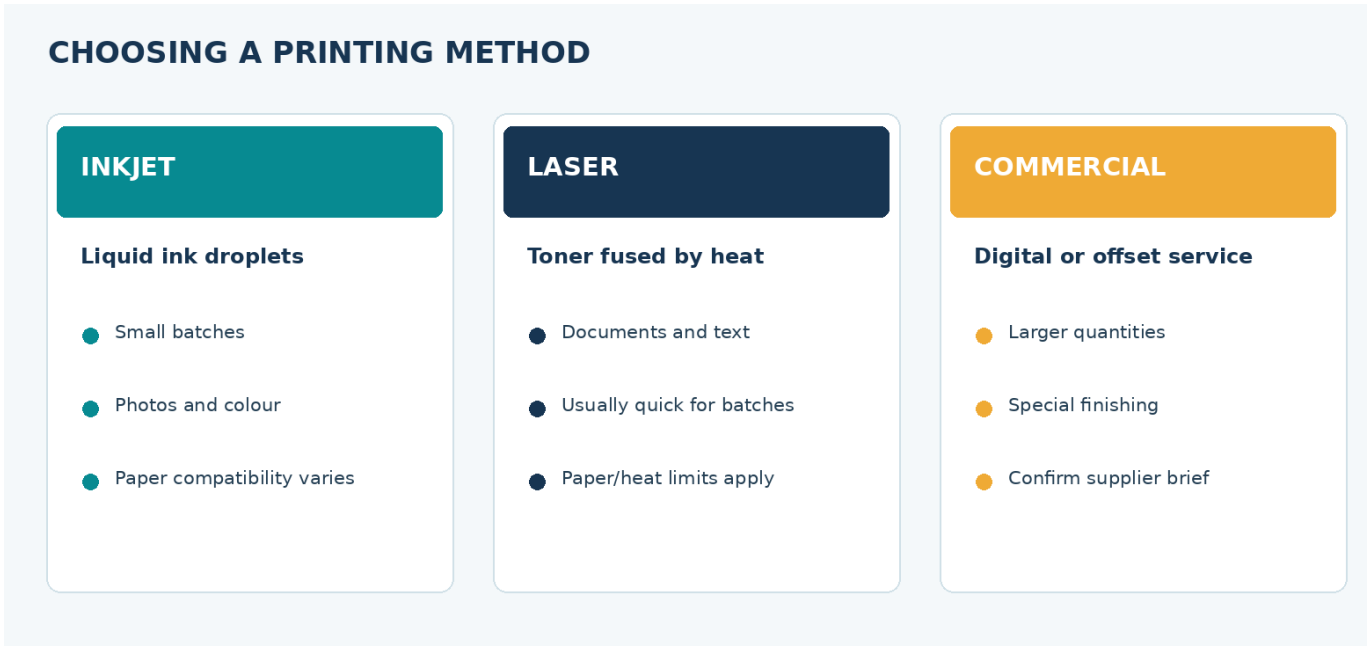
Printed publication

A physical hand-out requires appropriate paper, actual page dimensions, printer settings and acceptable image detail. Printed colours can differ from screen colours, so one test print is useful.

ILLUSTRATION

Which printing method should we choose?

There is no single printer type that is best for all purposes.



Compare needs

Inkjet may suit small colour image tasks; laser often suits repeated office text; commercial printing helps with special paper, larger runs and finishing. Costs, quality and availability vary by device and service.

KEY CONCEPT

Inkjet printing

Many small printers use inkjet technology.

Basic idea

An inkjet printer places droplets of liquid ink onto the paper. It may produce detailed photographic results when suitable ink and paper are used. However, drying, ink consumption and colour quality depend on the device and media.

Example: printing a colourful educational poster in a small office.

KEY CONCEPT

Laser printing

Often used for office text and repeated pages.

Basic idea

A laser printer typically transfers toner to paper and fixes it with heat. It can produce crisp text and efficient document printing, though colour quality and accepted paper types vary between models.

Example: producing a stack of black-and-white class hand-outs.

KEY CONCEPT

Commercial printing services

A print shop may offer more than a standard desktop printer.

Basic idea

Commercial printers can use digital presses or offset processes and offer specialised paper, folding, trimming and binding. Quantity, colour requirements, finishing and production time affect the method selected.

For a large order, ask the provider for page size, bleed, file format and finishing requirements before export.

EXPLAIN

Paper properties

The sheet itself changes the final result.

Size and thickness

A4 measures 210 × 297 mm. Paper weight is often expressed in gsm (grams per square metre). Ordinary office paper and heavier brochure stock are not interchangeable for every printer or purpose.

Finish and suitability

Coated and uncoated surfaces absorb or reflect ink differently. Some printers restrict coated, textured or heavy paper. Always use paper compatible with the specific printer and expected output.

Choose a suitable method

Think about the audience, quantity and resources.

YOUR TURN

Which is more sensible for 20 text-heavy course hand-outs: a local office laser printer or an offset press? Give two reasons.

Mention quantity, setup needs, paper and access; the answer can depend on real costs.

KEY CONCEPT

What is a print-ready document?

A file can look correct on screen but still be unsuitable for a press.

Working definition

A print-ready document is a publication checked and prepared to meet the requirements of its intended printing process. It normally has the correct dimensions, complete text, sound graphics, suitable fonts and an appropriately exported file.

Different print providers have different requirements, so always confirm the actual specification.

EXPLAIN

Editable file versus PDF proof

These two documents have different responsibilities.

Editable source (.indd)

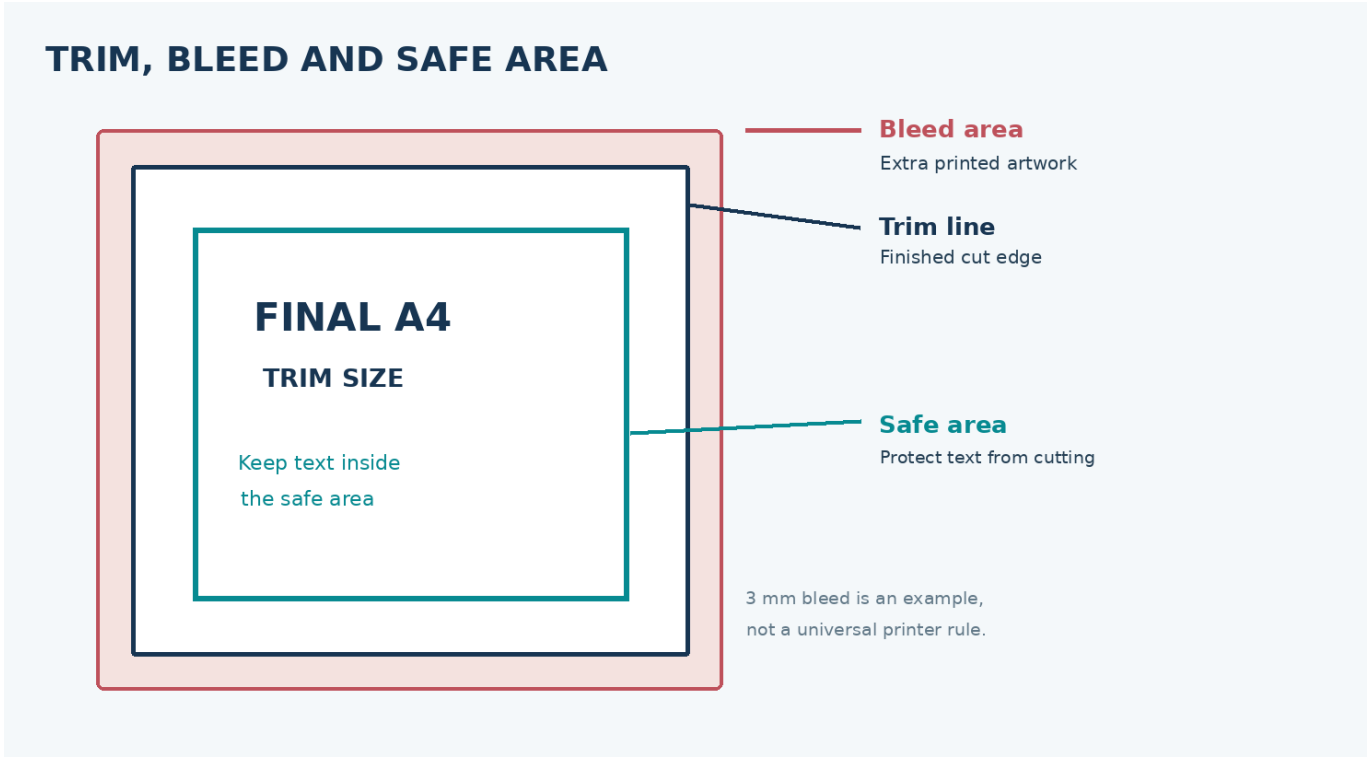
The InDesign file holds the editable layout and may depend on linked images and fonts. Keep it organised so text, positions and graphics can be corrected before final output.

PDF proof

The PDF shows the intended exported page for inspection and sharing. It is convenient for printing, but exporting does not automatically resolve hidden text, blurry images or an unsuitable page setup.

Trim, bleed and safe area

These zones should not be confused with each other.



Remember the difference

Trim is the intended finished edge. Bleed extends background artwork beyond that line. The safe area keeps essential information inside the cut. A 3 mm bleed is just an illustrative value: confirm the provider’s instruction.

KEY CONCEPT

Safe margins on an A4 hand-out

A normal office hand-out may not require a full commercial bleed setup.

Practical classroom choice

For our simple A4 hand-out, keep the title, body text and logo comfortably within the printable area. Margins protect readability and help avoid printer limitations. Background-to-edge printing requires a printer capable of borderless output or later trimming.

The supplied A4 PDF is an instructional example; it is not a certified commercial press file.

Printer dialogue: five settings to inspect

Menu names and available choices differ by device.

BEFORE PRESSING PRINT, CHECK THE SETTINGS

PAPER	**A4 (210 × 297 mm)**	Confirm output and feed orientation
SCALING	**Actual size / 100%**	Do not shrink without a reason
COLOUR	**Colour or greyscale**	Choose by purpose and printer
SIDES	**Single / double sided**	For one-page handout: single sided
QUALITY	**Draft / normal / best**	Use test results and paper type

Checklist

Confirm paper size, actual size/scale, sides, colour or greyscale, and output quality. Check the paper tray and orientation too. Never assume the software has chosen the right values automatically.

EXPLAIN

“Actual size” versus “Fit to page”

A small selection can change the physical dimensions.

Actual size (100%)

Normally prints at the intended scale when page and paper agree. Choose this for our A4 classroom proof unless the printer or document indicates otherwise.

Fit to page

May shrink or scale the image to a printable area. Useful in some situations, but it can alter measurements and reduce the size of text or layout. Inspect the preview before proceeding.

KEY CONCEPT

Orientation, paper and number of copies

Basic settings are as important as advanced options.

Practical rule

Choose A4 portrait for the Tumaini hand-out. Confirm the tray paper matches the PDF size and choose the number of copies needed. A single test copy is usually better than discovering a mistake after producing a large batch.

If a printer does not support edge-to-edge A4, the printable margins may affect output.

KEY CONCEPT

What does duplex mean?

Some documents use both sides of the paper.

Definition

Duplex printing places content on the front and back of the same sheet. It can reduce paper consumption for multipage documents. The long-edge or short-edge flip setting determines page orientation when turning the sheet.

Our one-page hand-out is normally single-sided; duplex is introduced for later publications.

Choose settings for one page

The PDF is already a single-page A4 portrait file.

YOUR TURN

State appropriate settings for paper size, scale, sides, colour and copies when producing ONE proof of the hand-out.

Use A4, actual size, single-sided and one copy; colour depends on the available printer.

KEY FORMULA

What does PPI mean?

Image detail has a relationship with its final printed size.

Effective PPI

Pixels per inch measures the density of image pixels at the size they are printed. The same photograph has lower effective PPI when placed at a larger physical width. This is different from the printer’s dots-per-inch (DPI) specification.

Effective PPI = (pixel width × 25.4) ÷ printed width in millimetres.

WORKED EXAMPLE

Calculate effective PPI

The image is 1,800 pixels wide and prints at 150 mm.

EFFECTIVE IMAGE RESOLUTION

IMAGE SOURCE

1,800 pixels wide

Placed at 150 mm wide

PPI = (1,800 × 25.4) ÷ 150

= 304.8 pixels per inch

WHAT IT MEANS

A larger printed width lowers PPI.

Changing the DPI setting alone

does not create missing detail.

Check actual size in the layout.

Worked solution

1,800 × 25.4 ÷ 150 = 304.8 PPI. This value is useful evidence about image detail at the intended size. The appropriate resolution depends on print technology, image type and viewing distance.

EXPLAIN

PPI is not the same as DPI

Two important terms often appear in print settings.

Image PPI

Describes how the image’s original pixels are distributed at the placed print dimensions. Enlarging an image usually lowers its effective PPI.

Printer DPI

Describes the density or addressability of printing marks. Selecting a higher DPI mode in the driver does not magically recreate details absent from the original image.

Your turn: PPI calculation

Show working before deciding whether the image is suitable.

YOUR TURN

A photograph is 1,200 pixels wide and is printed 150 mm wide. Calculate its effective PPI.

Use pixel width × 25.4 ÷ printed width in millimetres.

KEY CONCEPT

What should a good print look like?

A visual quality check considers more than colour alone.

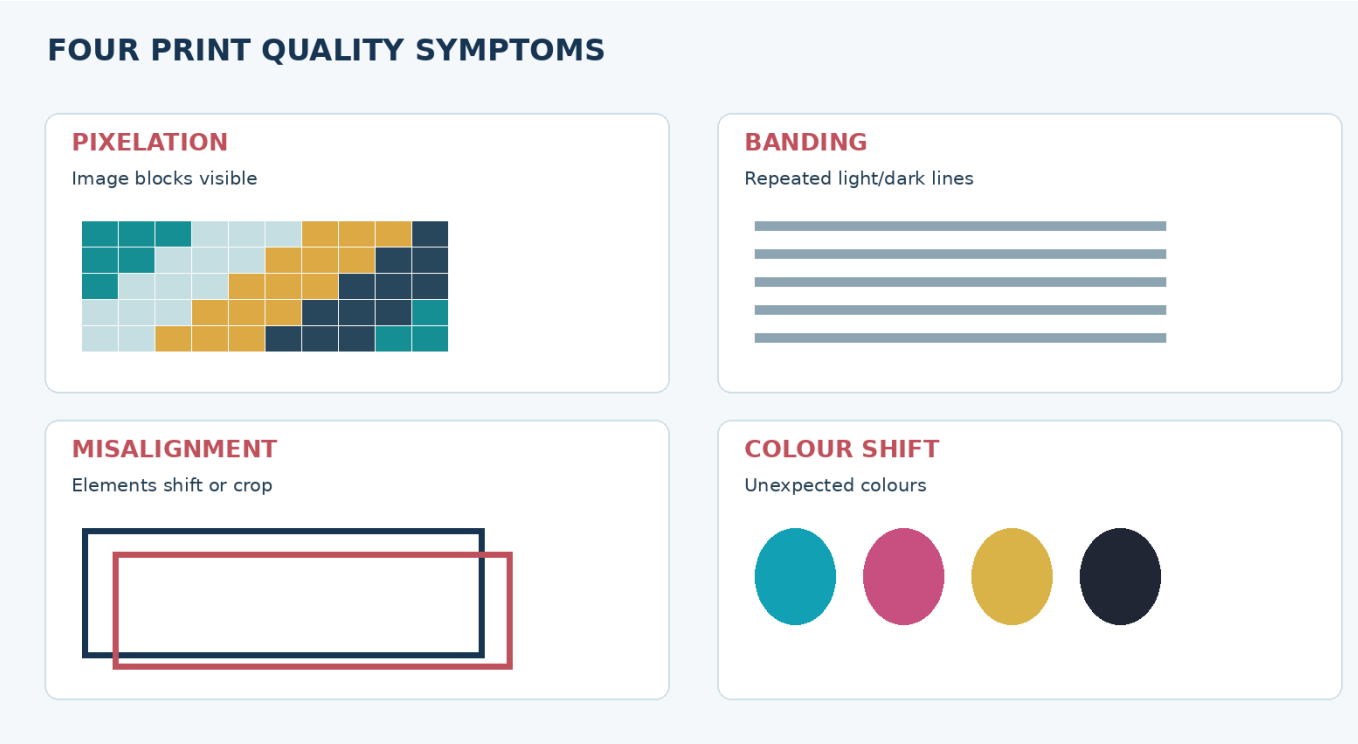
Inspect these properties

Printed body text should be sharp and legible. Images should contain sufficient detail. Content should be correctly placed and not clipped. Colour, streaking, margins and page orientation should be sensible for the specified output.

Compare the printed copy with the PDF proof before printing multiple copies.

Recognise common printing defects

A symptom is a starting point for diagnosis, not proof of one cause.



Four typical symptoms

Pixelation, banding, misalignment and colour shift can come from different problems. Compare the PDF with the print and check relevant settings before changing or replacing equipment.

KEY CONCEPT

Pixelation and blur

A photo that looks clear at small size may fail when enlarged.

Possible reasons

Pixelation can occur when too few pixels are spread across a large printed area. Blur can result from low detail, image editing or output conditions. Begin by checking the original photograph and effective PPI at the final page size.

Avoid enlarging a small screenshot or thumbnail to fill a large hand-out.

KEY CONCEPT

Banding and streaks

These marks may not exist in the original PDF.

Possible reasons

Banding consists of unwanted repeated light, dark or coloured lines. Causes may include printhead or toner issues, feed problems, media and maintenance needs. Compare the source with a print test and use the manufacturer’s prescribed checks.

Do not assume all banding can be fixed by changing image contrast.

KEY CONCEPT

Misalignment and unexpected cropping

Paper size and driver settings are common places to investigate.

Possible reasons

If content is shifted, cut off or reduced, compare PDF page size, paper orientation, scaling and available printer margins. If the PDF itself is already incorrect, return to the editable publication instead of repeatedly adjusting the printer.

One-page test prints help distinguish document errors from printer-setting problems.

KEY CONCEPT

Colour differences between screen and print

Screens and paper produce colour in different ways.

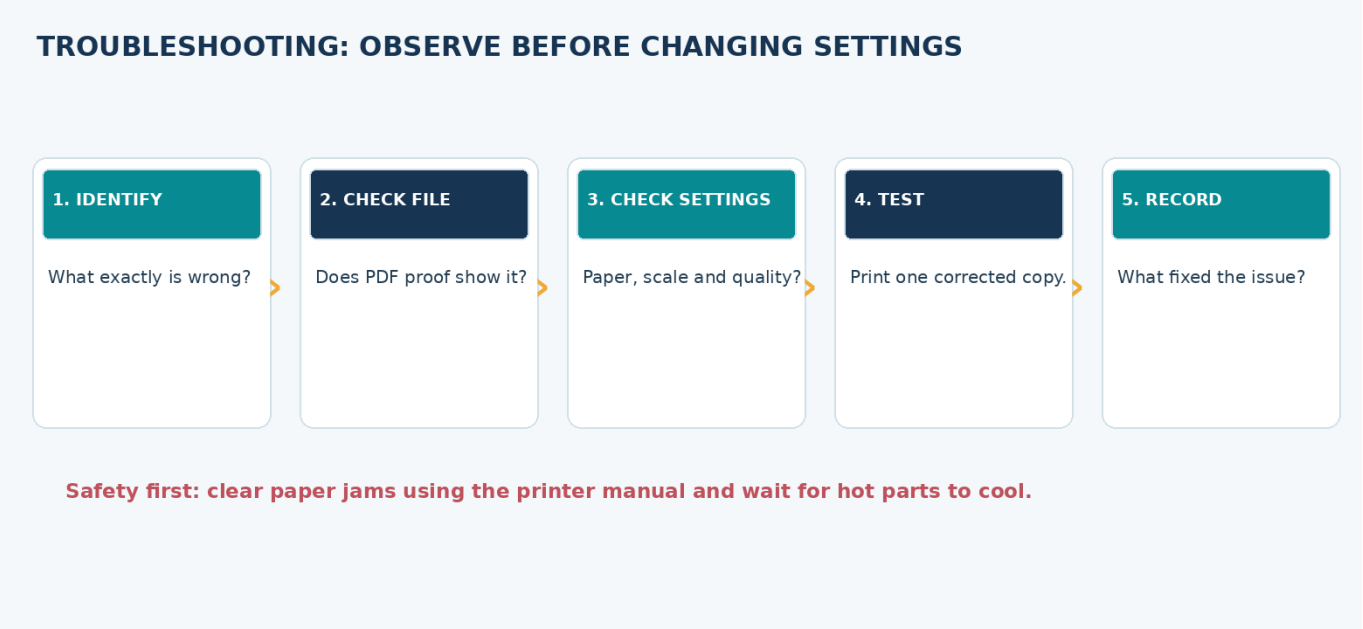
Possible reasons

Screen brightness, colour profiles, ink or toner, media type and printer condition can all affect colour appearance. Print may look less vivid than the monitor. For a classroom task, prioritise readable contrast and consistent, intentional use of colour.

For professional accuracy, use the print provider’s colour specifications and proofing process.

Safe diagnosis is a step-by-step process

Avoid experimenting with the whole printer at once.



Five checks

Name the symptom; inspect the PDF; verify size, media and settings; print one corrected sample; and record what changed. If hardware work is needed, follow the manufacturer’s safety guidance.

Paper jams and printer maintenance

Students should know basic precautions, not internal repair.

Safety first

Power, heat and moving rollers can make printer maintenance hazardous. If a sheet jams, stop the job and follow the printer manual or lab technician’s instructions. Avoid forcing paper through the machine or touching hot internal parts.

When in doubt, report the problem instead of dismantling the equipment.

COST EXAMPLE

Estimate simple printing costs

Only use a quoted price when planning real jobs.

Basic method

The number of required sheets depends on page count, copies and whether printing is single- or double-sided. Multiply the number of printed sheets by the stated unit price; add other supplier charges if applicable. Classroom figures are hypothetical.

Example: 30 single-page hand-outs × TZS 150 per printed sheet = TZS 4,500.

Quick cost calculation

Use a fictional teaching quotation.

YOUR TURN

A training centre needs 40 one-page, single-sided A4 hand-outs at TZS 120 per sheet. What is the estimated charge, excluding other costs?

Multiply 40 × 120, and label your answer in Tanzanian shillings.

Practical: prepare and test the hand-out

Use the completed “Stay Safe Online” PDF from Lecture 8.

1	**Inspect PDF**	Check title, tips, placed graphic and A4 page size.
2	**Choose settings**	Select paper, scale, colour and number of copies.
3	**Calculate**	Find effective PPI and estimated printed-copy cost.
4	**Print or analyse**	Use a test page if available; otherwise inspect defect examples.
5	**Record**	Write observations and a justified correction or next step.

The test page is a diagnostic aid

An A4 chart is supplied in the practical resources.

How to use it

Print the chart at actual size where possible. Compare large and small text, coloured areas, fine lines and page alignment. Record what you notice. The chart can support a basic classroom quality check but is not a professional colour-calibration tool.

If no printer is available, use the printed-defect illustrations and complete a reasoned diagnosis.

Diagnose a problem

Decide which part of the workflow should be checked first.

YOUR TURN

The hand-out looks correct as a PDF, but the printed page is unexpectedly smaller. Name one likely cause and one appropriate correction.

Check scaling and paper size before changing the design itself.

KEY CONCEPT

Keep a useful quality-control record

A short record helps explain why a decision was made.

Record evidence

Write down the file name, printer settings, paper size, number of copies and observed problem. State the suspected cause and what you checked. A useful report separates what was actually observed from what was only assumed.

Example: “PDF is A4; driver used Fit to page; test output is smaller; retry at 100%.”

Five things to remember

These ideas connect the entire lecture.

1	**Choose**	A printer and paper appropriate for the publication.
2	**Verify**	PDF size, margins and any provider requirements.
3	**Calculate**	Effective PPI at the actual placed size.
4	**Inspect**	One sample for clarity, colour and positioning.
5	**Correct**	The source or settings, then test again before bulk printing.

End-of-lecture review

Think about the physical output, not only the PDF.

YOUR TURN

Why should you inspect a test print before producing 100 copies? State three different problems that a test may reveal.

Examples include size, image sharpness, colour, cropping and streaks.

COMING NEXT

Next lecture: Image Processing and Compression

Efficient digital images balance file size and visual quality.

Connection to Lecture 10

You now know how image resolution and printing size affect output quality. In the next lecture, you will compare lossless and lossy formats, measure changes in file size and decide how to preserve useful image quality for digital and printed media.

Keep your Tumaini image files for the next practical.